

Motion planning

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Task and Motion Planning

Theoretical concepts

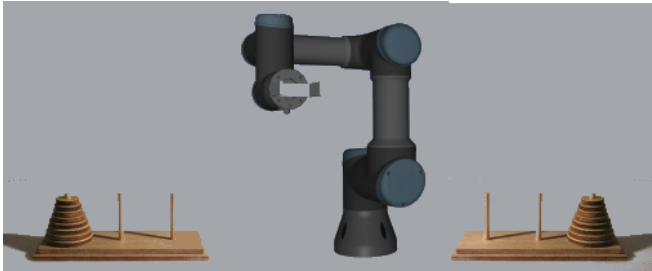
Implementation

Task and Motion Planning

Theoretical concepts

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Definition

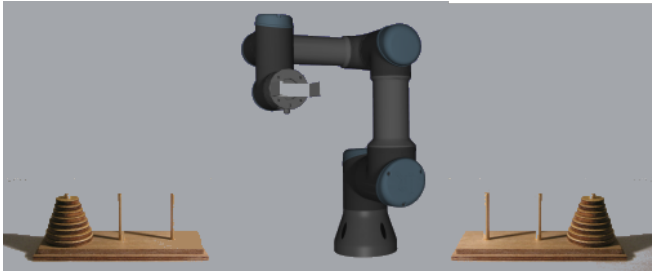


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- ▶ One or several objects,
- ▶ initial configurations for robots and objects
- ▶ goal configurations for robots and objects

Task and Motion planning : automatically computing a feasible trajectory between the initial and goal configurations.

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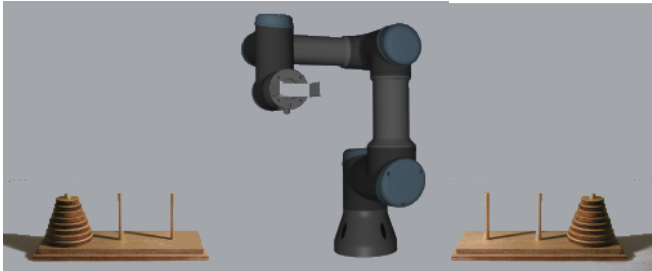


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Configuration space

Configuration

$$\mathbf{q} = (\mathbf{q}_{r_1}, \mathbf{q}_{r_2}, \mathbf{q}_{c_1}, \mathbf{q}_{c_2}, \mathbf{q}_{s_1}, \mathbf{q}_{s_2})$$

$$\mathbf{q}_{r_1}, \mathbf{q}_{r_2} \in \mathbb{R}^6$$

$$\mathbf{q}_{c_1}, \mathbf{q}_{c_2}, \mathbf{q}_{s_1}, \mathbf{q}_{s_2} \in \mathbb{R}^7$$

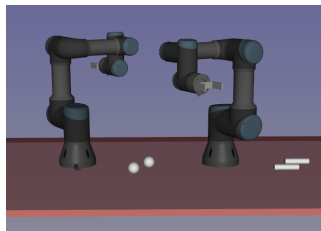
where

$\mathbf{q}_{r_1} = (q_1, \dots, q_6)$ is the vector of joint angles,

$\mathbf{q}_{c_1} = (x, y, z, X, Y, Z, W)$

$W + Xi + Yj + Zk$ is a **unit** quaternion.

The configuration space is a differential manifold.



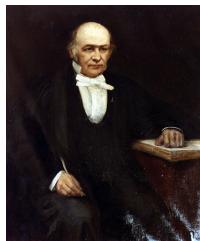
Quaternions

Non-commutative field isomorphic to $\mathbb{R}^4$, spanned by three elements i, j, k that satisfy the following relations :

$$i^2 = j^2 = k^2 = ijk = -1$$

from which we immediately deduce

$$ij = k, \quad jk = i, \quad ki = j$$



Hamilton (1843)

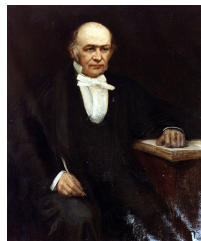
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Hamilton (1843)

Unit Quaternions and rotations

Let $q = q_0 + q_1i + q_2j + q_3k$ be a unit quaternion :

$$q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1$$

$\forall x = (x_0, x_1, x_2) \in \mathbb{R}^3$, let $u = x_0i + x_1j + x_2k$

$$q \cdot u \cdot q^* = y_0i + y_1j + y_2k$$

where $q^* = q_0 - q_1i - q_2j - q_3k$ is the conjugate of q .

$y = (y_0, y_1, y_2)$ is the image of x by the rotation of matrix

$$\begin{pmatrix} 1 - 2(q_2^2 + q_3^2) & 2q_2q_1 - 2q_3q_0 & 2q_3q_1 + 2q_2q_0 \\ 2q_2q_1 + 2q_3q_0 & 1 - 2(q_1^2 + q_3^2) & 2q_3q_2 - 2q_1q_0 \\ 2q_3q_1 - 2q_2q_0 & 2q_3q_2 + 2q_1q_0 & 1 - 2(q_1^2 + q_2^2) \end{pmatrix}$$

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Unit Quaternions and rotations

- ▶ Notice that q and $-q$ represent the same rotation

Definitions

- ▶ Workspace : $\mathcal{W} = \mathbb{R}^2$ or $\mathbb{R}^3$: space in which the robot evolves
- ▶ Obstacle in workspace : compact subset of $\mathcal{W}$, denoted by $\mathcal{O}$.
- ▶ Configuration space : $\mathcal{C}$.
- ▶ Position in configuration $\mathbf{q}$ of a point $M \in \mathcal{B}_i$: $\mathbf{x}_i(M, \mathbf{q})$.
- ▶ Obstacle in the configuration space :

$$\begin{aligned}\mathcal{C}_{obst} = \{ \mathbf{q} \in \mathcal{C}, \quad & \exists i \in \{1, \dots, m\}, \exists M \in \mathcal{B}_i, \mathbf{x}_i(M, \mathbf{q}) \in \mathcal{O} \text{ or} \\ & \exists i, j \in \{1, \dots, m\}, \exists M_i \in \mathcal{B}_i, \exists M_j \in \mathcal{B}_j, \\ & \mathbf{x}_i(M_i, \mathbf{q}) = \mathbf{x}_j(M_j, \mathbf{q}) \} \end{aligned}$$

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Motion

- ▶ Configuration space :
 - ▶ differential manifold
- ▶ Motion :
 - ▶ continuous function from $[0, 1]$ to $\mathcal{C}$.
- ▶ Collision-free motion :
 - ▶ continuous function from $[0, 1]$ to $\mathcal{C}_{free}$.

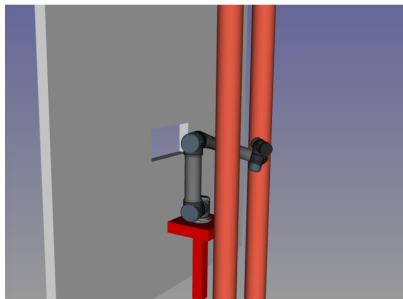
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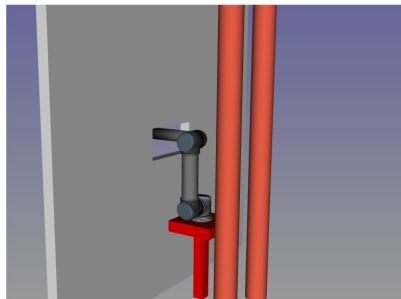
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Motion planning problem



initial configuration



goal configuration

$$\mathcal{C} = \mathbb{R}^6$$

History

- ▶ Before the 1990's : mainly a mathematical problem
 - ▶ real algebraic geometry,
 - ▶ decidability : Schwartz and Sharir 1982,
 - ▶ Tarski theorem, Collins decomposition,
- ▶ from the 1990's : an algorithmic problem
 - ▶ random sampling (1993),
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Random methods

- ▶ In the early 1990's, random methods started being developed
- ▶ Principle
 - ▶ shoot random configurations
 - ▶ test whether they are in collision
 - ▶ build a graph (roadmap) the nodes of which are free configurations
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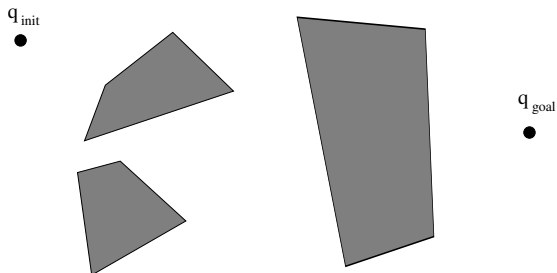
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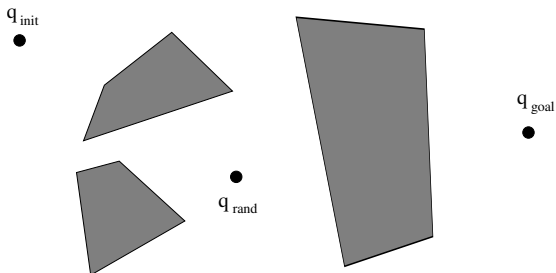
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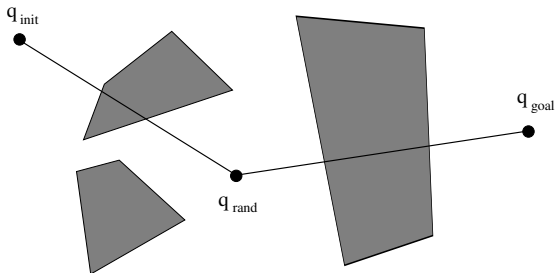
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Pick a random configuration



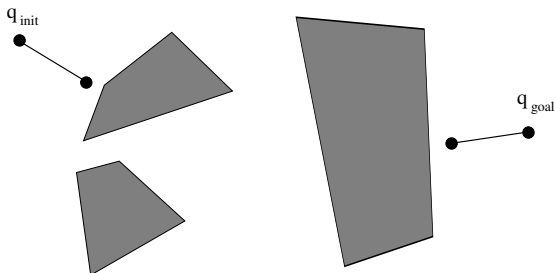
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Try to connect it to the nearest nodes of each connected component



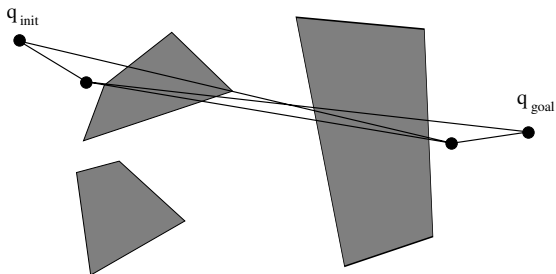
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Keep collision-free parts of paths

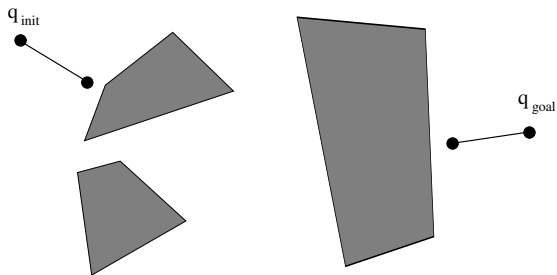


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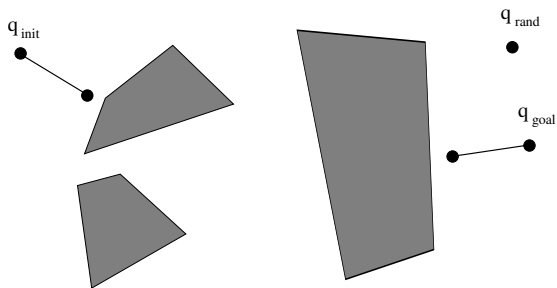
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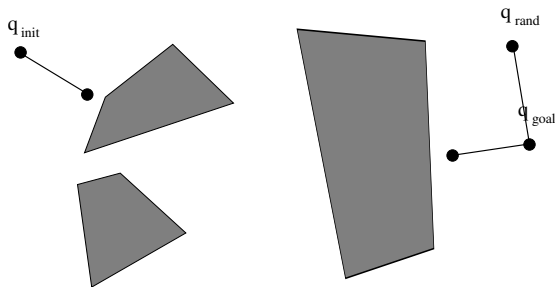
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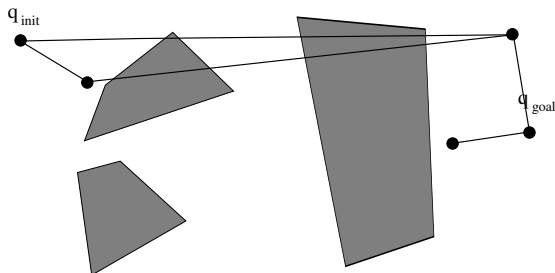
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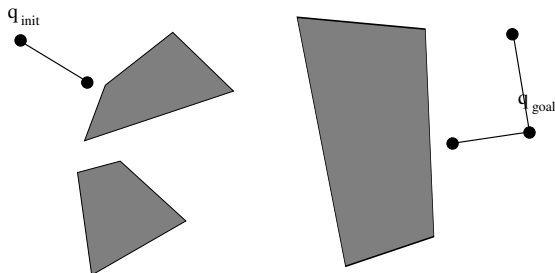
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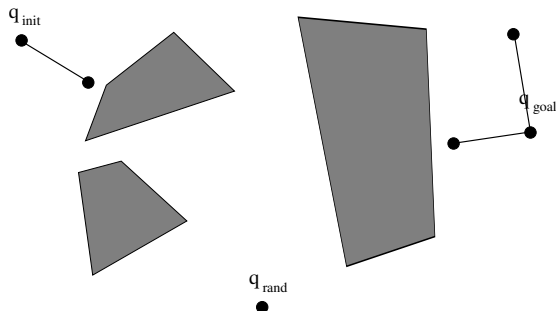
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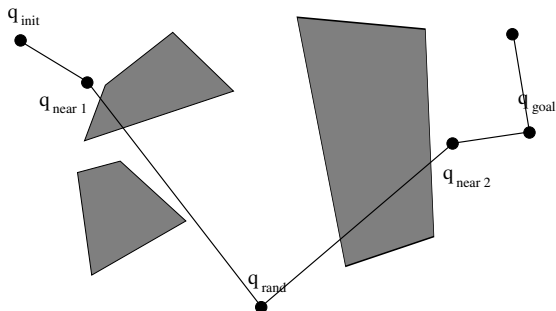
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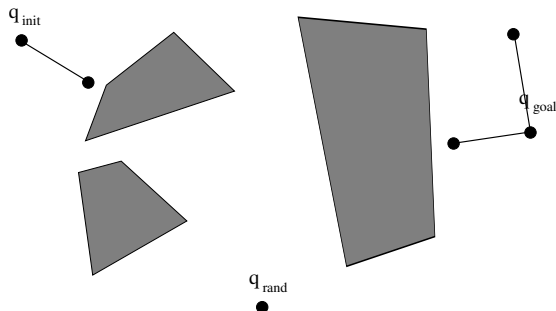
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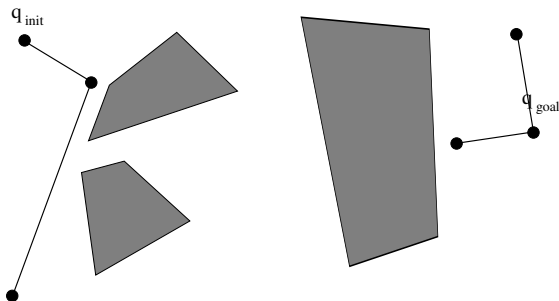
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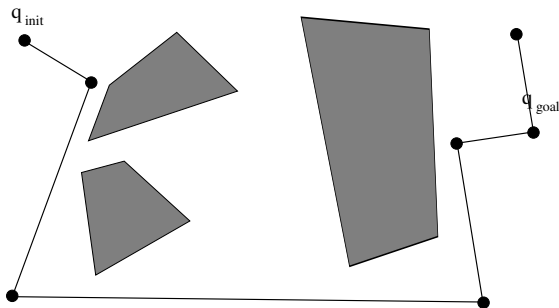
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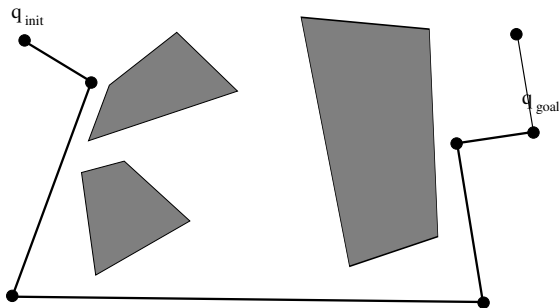
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- ▶ Pros :
 - ▶ no explicit computation of the free configuration space,
 - ▶ easy to implement,
 - ▶ robust.
- ▶ Cons :
 - ▶ no completeness property, only probabilistic completeness,
 - ▶ difficult to find narrow passages.
- ▶ Requested operators :
 - ▶ Collision tests
 - ▶ for configurations (static),
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A manipulation motion

- ▶ is the motion of
 - ▶ one or several robots and of
 - ▶ one or several objects
- ▶ such that each object
 - ▶ either is in a stable position, or
 - ▶ is moved by one or several robots.

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Numerical constraints

Constraints to which manipulation motions are subject can be expressed numerically.

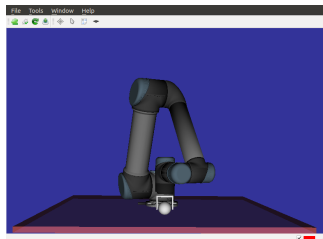
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$$f(\mathbf{q}) = 0, \quad \begin{array}{l} m \in \mathbb{N}, \\ f \in C^1(\mathcal{C}, \mathbb{R}^m) \end{array}$$

- ▶ Parameterizable numerical constraints :

$$f(\mathbf{q}) = f_0, \quad \begin{array}{l} m \in \mathbb{N}, \\ f \in C^1(\mathcal{C}, \mathbb{R}^m) \\ f_0 \in \mathbb{R}^m \end{array}$$

Example : robot manipulating a ball



$$\mathcal{C} = [-\pi, \pi]^6 \times \mathbb{R}^3 \quad (1)$$

$$\mathbf{q} = (q_0, \dots, q_5, x_b, y_b, z_b) \quad (2)$$

Two *states* :

- ▶ placement : the ball is lying on the table,
- ▶ grasp : the ball is held by the end-effector.

Example : robot manipulating a ball

Each state is defined by a numerical constraint

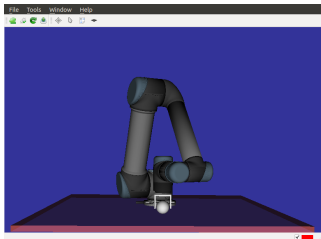
► placement

$$z_b = 0$$

► grasp

$$\mathbf{x}_{gripper}(q_0, \dots, q_5) - \begin{pmatrix} x_b \\ y_b \\ z_b \end{pmatrix} = 0$$

Each state is a sub-manifold of the configuration space



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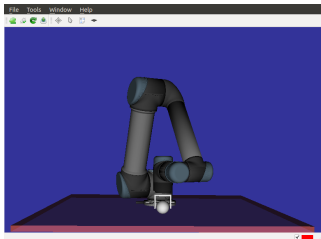
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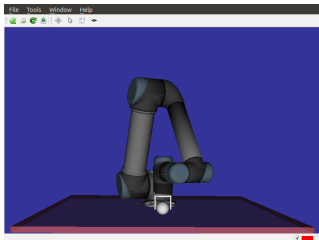
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Motion constraints



Two *types of motion* :

- ▶ transit : the ball is lying and **fixed** on the table,
- ▶ transfer : the ball moves with the end-effector.

Example : robot manipulating a ball

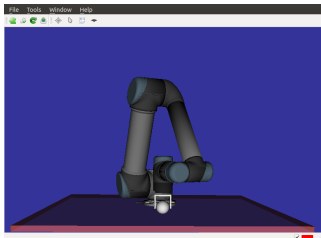
Motion constraints

► transit

$$\begin{array}{ll} x_b = x_0 & \} \text{ parameterizable} \\ y_b = y_0 & \\ z_b = 0 & \} \text{ simple} \end{array}$$

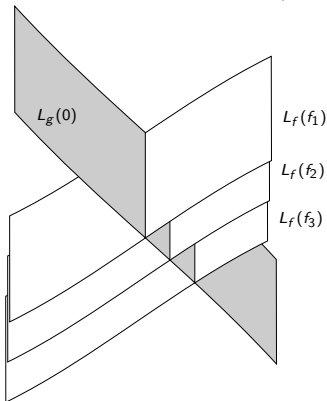
► transfer

$$\mathbf{x}_{gripper}(q_0, \dots, q_5) - \begin{pmatrix} x_b \\ y_b \\ z_b \end{pmatrix} = 0$$



Foliation

Motion constraints define a foliation of the admissible configuration space ($\text{grasp} \cup \text{placement}$).



- ▶ f : position of the ball

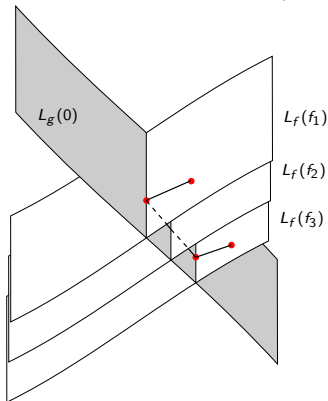
$$L_f(f_1) = \{\mathbf{q} \in \mathcal{C}, f(\mathbf{q}) = f_1\}$$

- ▶ g : grasp of the ball

$$L_g(0) = \{\mathbf{q} \in \mathcal{C}, g(\mathbf{q}) = 0\}$$

Foliation

Motion constraints define a foliation of the admissible configuration space ($\text{grasp} \cup \text{placement}$).



Solution to a manipulation planning problem is a concatenation of *transit* and *transfer* paths.

General case

In a manipulation problem,

- ▶ the state of the system is subject to
 - ▶ numerical constraints
- ▶ trajectories of the system are subject to
 - ▶ numerical constraints
 - ▶ parameterizable numerical constraints.

General case

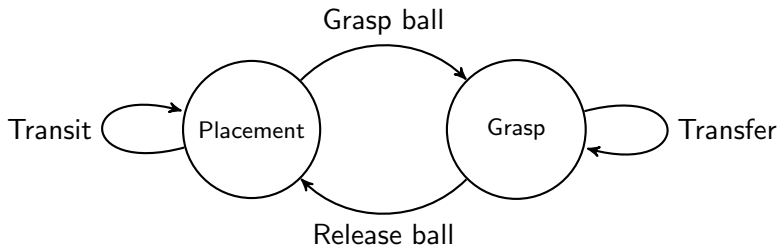
In a manipulation problem,

- ▶ the state of the system is subject to
 - ▶ numerical constraints
- ▶ trajectories of the system are subject to
 - ▶ numerical constraints
 - ▶ parameterizable numerical constraints.

Constraint graph

A manipulation planning problem can be represented by a *manipulation graph*.

- ▶ **Nodes** or *states* are numerical constraints.
- ▶ **Edges** or *transitions* are parameterizable numerical constraints.



Projecting configuration on constraint

Newton-Raphson algorithm

- ▶ $\mathbf{q}_0$ configuration,
- ▶ $f(\mathbf{q}) = 0$ non-linear constraint,
- ▶ ϵ numerical tolerance

Projection $(\mathbf{q}_0, f)$:

$\mathbf{q} = \mathbf{q}_0 ; \alpha = 0.95$

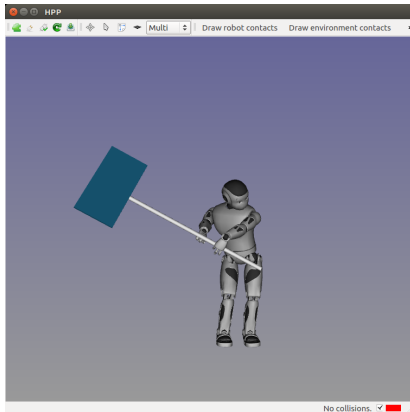
for i from 1 to max_iter :

$$\mathbf{q} = \mathbf{q} - \alpha \left(\frac{\partial f}{\partial \mathbf{q}}(\mathbf{q}) \right)^+ f(\mathbf{q})$$

if $\|f(\mathbf{q})\| < \epsilon$: return $\mathbf{q}$

return failure

Newton-Raphson algorithm

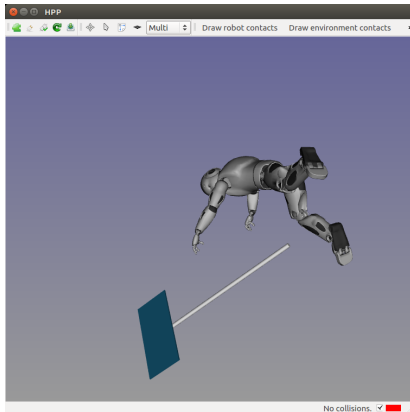


Goal : Generate a configuration satisfying the constraints.

Constraints

- ▶ quasi-static equilibrium (15)
- ▶ both hands hold the placard (10)

Newton-Raphson algorithm

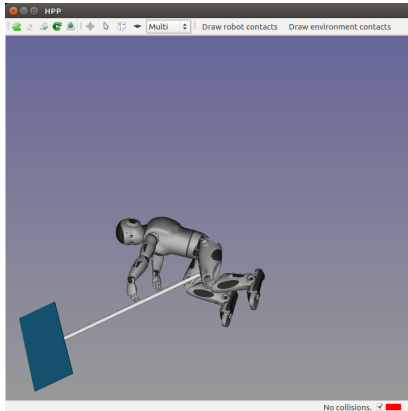


Shoot random configuration

Constraints

- ▶ quasi-static equilibrium (15)
- ▶ both hands hold the placard (10)

Newton-Raphson algorithm

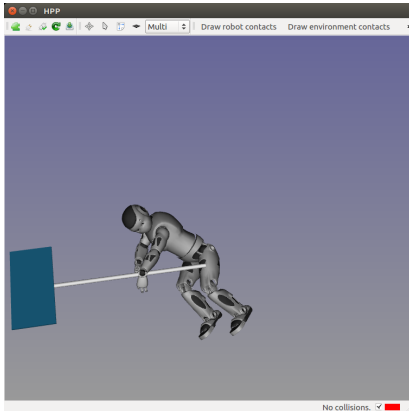


Solve linearized system

Constraints

- ▶ quasi-static equilibrium (15)
- ▶ both hands hold the placard (10)

Newton-Raphson algorithm

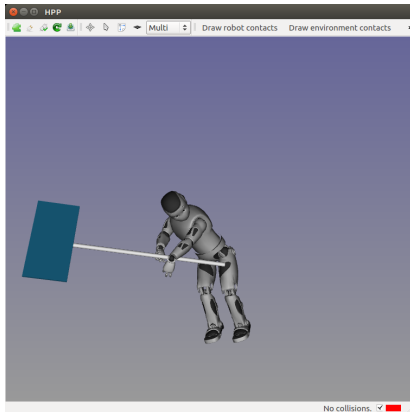


Solve linearized system

Constraints

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Newton-Raphson algorithm

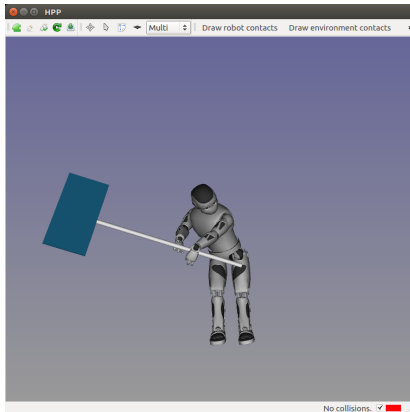


Solve linearized system

Constraints

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Newton-Raphson algorithm

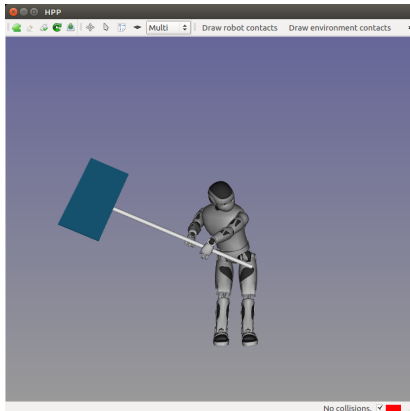


Solve linearized system

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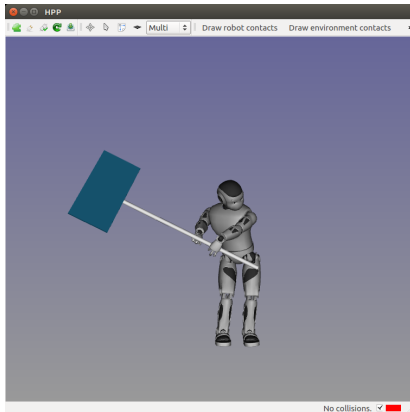


Solve linearized system

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Newton-Raphson algorithm

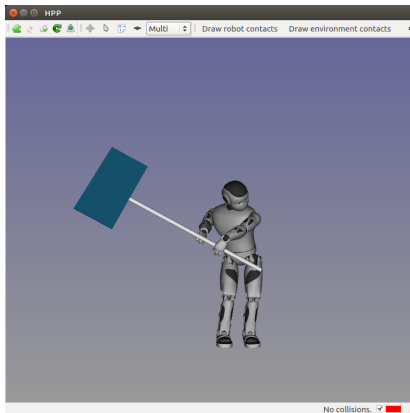


Solve linearized system

Constraints

- ▶ quasi-static equilibrium (15)
- ▶ both hands hold the placard (10)

Newton-Raphson algorithm



Constraints

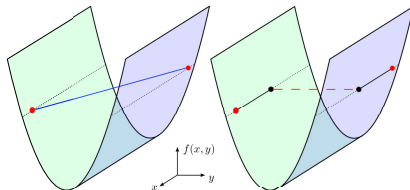
- ▶ quasi-static equilibrium (15)
- ▶ both hands hold the placard (10)

Result : a configuration that satisfies the constraints (up to given threshold).

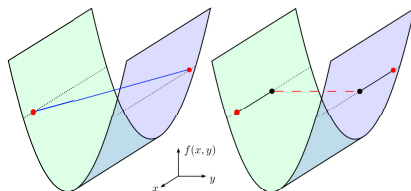
Projecting path on constraint

- ▶ $path$: mapping from $[0, 1]$ to $\mathcal{C}$
- ▶ $f(\mathbf{q}) = 0$ non-linear constraint,

Applying Newton Raphson at each point may result in a discontinuous path



Discontinuous Projection



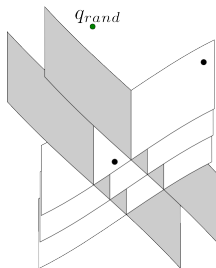
$$\mathcal{C} = \mathbb{R}^2, f(x, y) = y^2 - 1$$

$$\frac{\partial f}{\partial \mathbf{q}} = \begin{pmatrix} 0 & 2y \end{pmatrix}, \quad \frac{\partial f^+}{\partial \mathbf{q}} = \begin{pmatrix} 0 \\ \frac{1}{2y} \end{pmatrix} \text{ ou } \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$y_{i+1} = y_i + \frac{1 - y_i^2}{2y_i}$$

Algorithm

Manipulation RRT



Manipulation RRT

$\mathbf{q}_{rand} = \text{shoot_random_config}()$

for each connected component :

```

 $\mathbf{q}_{near} = \text{nearest\_neighbor}(\mathbf{q}_{rand}, \text{roadmap})$ 
 $T = \text{select\_transition}(\mathbf{q}_{near})$ 
 $\mathbf{q}_{proj} = \text{generate\_target\_config}(\mathbf{q}_{near}, \mathbf{q}_{rand}, T)$ 
 $\mathbf{q}_{new} = \text{extend}(\mathbf{q}_{near}, \mathbf{q}_{proj}, T)$ 
 $\text{roadmap.insert\_node}(\mathbf{q}_{new})$ 
 $\text{roadmap.insert\_edge}(T, \mathbf{q}_{near}, \mathbf{q}_{new})$ 
 $\text{new\_nodes.append}(\mathbf{q}_{new})$ 

```

for $\mathbf{q} \in (\mathbf{q}_{new}^1, \dots, \mathbf{q}_{new}^{n_{cc}})$:

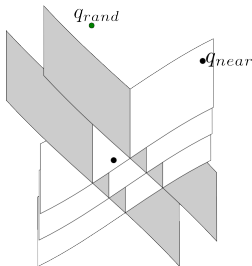
```

connect ( $\mathbf{q}$ , roadmap)

```

Algorithm

Manipulation RRT



Manipulation RRT

$\mathbf{q}_{rand} = \text{shoot_random_config}()$

for each connected component :

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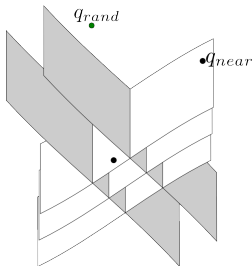
$\text{new_nodes.append}(\mathbf{q}_{new})$

for $\mathbf{q} \in (\mathbf{q}_{new}^1, \dots, \mathbf{q}_{new}^{n_{cc}})$:

$\text{connect}(\mathbf{q}, \text{roadmap})$

Algorithm

Manipulation RRT



Manipulation RRT

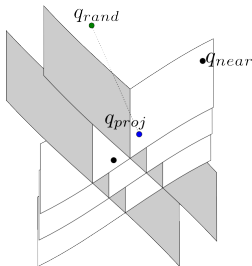
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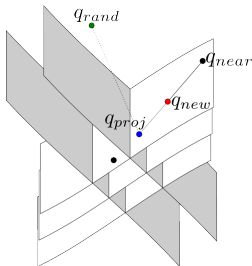
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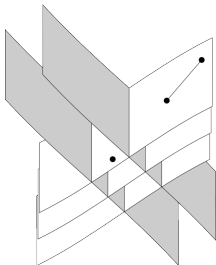
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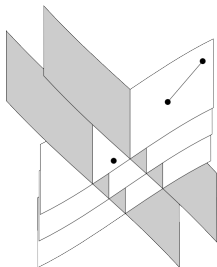
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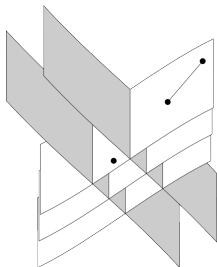
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Algorithm

Manipulation RRT



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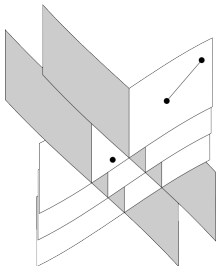
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Algorithm

Manipulation RRT



Manipulation RRT

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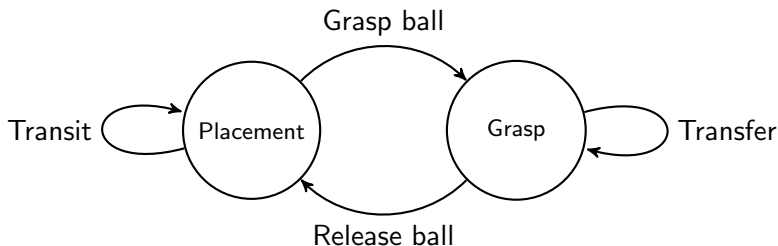
for $\mathbf{q} \in (\mathbf{q}_{new}^1, \dots, \mathbf{q}_{new}^{n_{cc}})$:

connect ($\mathbf{q}$, roadmap)

Select transition

$$T = \text{select_transition}(\mathbf{q}_{near})$$

Outward transitions of each state are given a probability distribution. The transition from a state to another state is chosen by random sampling.



Generate target configuration

$$\mathbf{q}_{proj} = \text{generate_target_config}(\mathbf{q}_{near}, \mathbf{q}_{rand}, T)$$

Once transition T has been selected, $\mathbf{q}_{rand}$ is *projected* onto the destination state S_{dest} in a configuration reachable by $\mathbf{q}_{near}$.

$$\begin{aligned} f_T(\mathbf{q}_{proj}) &= f_T(\mathbf{q}_{near}) \\ f_{S_{dest}}(\mathbf{q}_{proj}) &= 0 \end{aligned}$$

Extend

$$\mathbf{q}_{new} = \text{extend}(\mathbf{q}_{near}, \mathbf{q}_{proj}, T)$$

Project straight path $[\mathbf{q}_{near}, \mathbf{q}_{proj}]$ on transition constraint :

- ▶ if projection successful and projected path collision free

$$\mathbf{q}_{new} \leftarrow \mathbf{q}_{proj}$$

- ▶ otherwise $(\mathbf{q}_{near}, \mathbf{q}_{new}) \leftarrow$ largest path interval tested as collision-free with successful projection.

$$\forall \mathbf{q} \in (\mathbf{q}_{near}, \mathbf{q}_{new}), f_T(\mathbf{q}) = f_T(\mathbf{q}_{near})$$

Extend

$$\mathbf{q}_{new} = \text{extend}(\mathbf{q}_{near}, \mathbf{q}_{proj}, T)$$

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$$\forall \mathbf{q} \in (\mathbf{q}_{near}, \mathbf{q}_{new}), f_T(\mathbf{q}) = f_T(\mathbf{q}_{near})$$

Connect

connect ($\mathbf{q}$, roadmap)

for each connected component cc not containing $\mathbf{q}$:

for all n closest config $\mathbf{q}_1$ to $\mathbf{q}$ in cc :

▶ connect ($\mathbf{q}$, $\mathbf{q}_1$)

Connect

```
connect ( $\mathbf{q}_0$ ,  $\mathbf{q}_1$ ) :  
     $S_0$  = state ( $\mathbf{q}_0$ )  
     $S_1$  = state ( $\mathbf{q}_1$ )  
     $T$  = transition ( $S_0$ ,  $S_1$ )  
    if  $T$  and  $f_T(\mathbf{q}_0) == f_T(\mathbf{q}_1)$  :  
        if p = projected_path ( $T$ ,  $\mathbf{q}_0$ ,  $\mathbf{q}_1$ ) collision-free :  
            roadmap.insert_edge ( $T$ ,  $\mathbf{q}_0$ ,  $\mathbf{q}_1$ )  
    return
```

Relative positions as numerical constraints

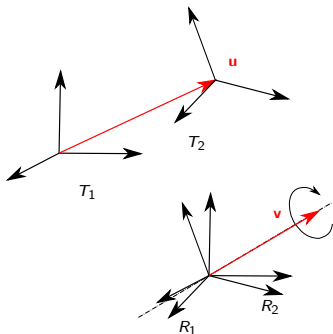
- ▶ $T_1 = T_{(R_1, t_1)} \in SE(3)$,
 $T_2 = T_{(R_2, t_2)} \in SE(3)$.
- ▶ $T_{2/1} = T_1^{-1} \circ T_2$ can be represented by a vector of dimension 6 :

$$\begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix}$$

where

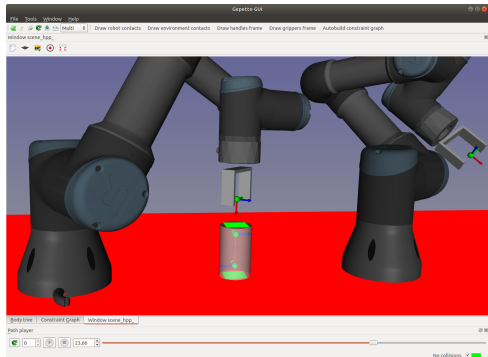
$$\mathbf{u} = R_1^T (t_2 - t_1)$$

$R_1^T R_2$ matrix of the rotation
around axis $\mathbf{v}/\|\mathbf{v}\|$ and of
angles $\|\mathbf{v}\|$.



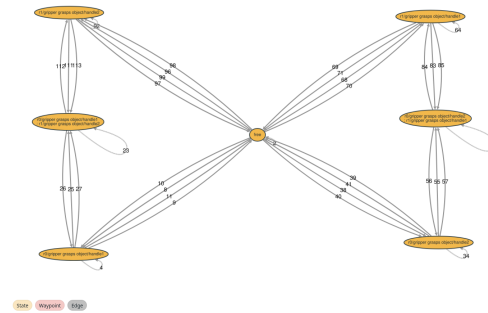
Definition of a manipulation planning problem

- ▶ grippers (frames attached to a joint),
 - ▶ handles (frames attached to a joint),
 - ▶ contact surfaces
- define grasps and stable poses for objects.

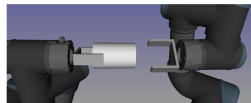
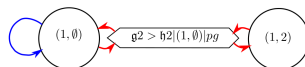


The factory

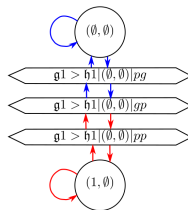
Given all the possible grasps (defined by grippers and handles), a factory builds the constraint graph with all possible states



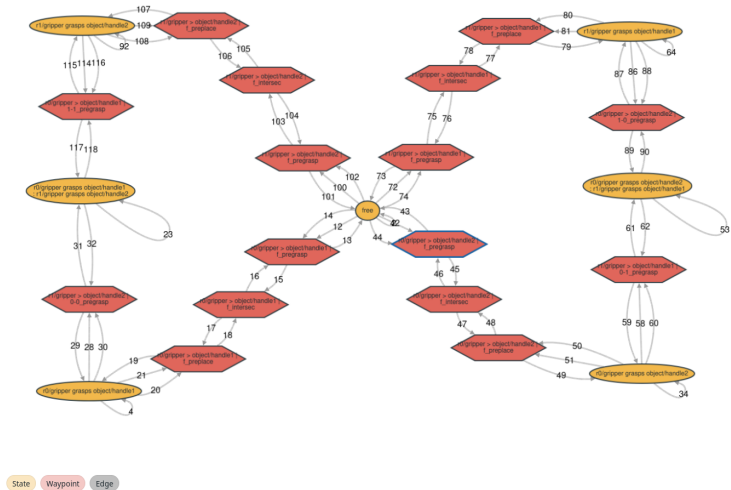
Waypoint transitions and states



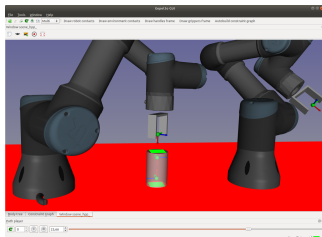
To ease manipulation planning resolution, some intermediate states are inserted by the factory.



Waypoint transitions and states



Motions on Lie groups



The configuration space of a manipulation problem, here $\mathcal{C} = \mathbb{R}^6 \times \mathbb{R}^6 \times SE(3)$

- ▶ is a Cartesian product of $\mathbb{R}^n$, $SO(2)$, $SO(3)$, $SE(2)$, $SE(3)$

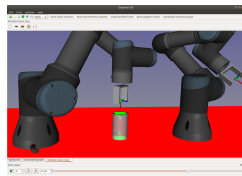
Each of those space share the properties of Lie groups.

Motions on Lie groups

The elements and derivatives of those space are represented by vectors.

Lie group type	configuration	velocity
$\mathbb{R}^n$	$\mathbf{q} \in \mathbb{R}^n$	$\mathbf{v} \in \mathbb{R}^n$
$SE(3)$	$(x_1, x_2, x_3, p_1, \dots, p_4) \in \mathbb{R}^7$	$(\mathbf{v}, \omega) \in \mathbb{R}^6$
$SE(2)$	$(x_1, x_2, \cos \theta, \sin \theta) \in \mathbb{R}^4$	$(\mathbf{v}, \dot{\theta}) \in \mathbb{R}^3$
$SO(3)$	$(p_1, p_2, p_3, p_4) \in \mathbb{R}^4$	$\omega \in \mathbb{R}^3$
$SO(2)$	$(\cos \theta, \sin \theta) \in \mathbb{R}^2$	$\dot{\theta} \in \mathbb{R}$

$$(q_1 \dots q_6 \ q_7 \dots q_{12} \times y \ z \ p_1 \ p_2 \ p_3 \ p_4)$$



Motions on Lie groups

Lie group type	configuration	velocity
$\mathbb{R}^n$	$\mathbf{q} \in \mathbb{R}^n$	$\mathbf{v} \in \mathbb{R}^n$
$SE(3)$	$(x_1, x_2, x_3, p_1, \dots, p_4) \in \mathbb{R}^7$	$(\mathbf{v}, \omega) \in \mathbb{R}^6$
$SE(2)$	$(x_1, x_2, \cos \theta, \sin \theta) \in \mathbb{R}^4$	$(\mathbf{v}, \dot{\theta}) \in \mathbb{R}^3$
$SO(3)$	$(p_1, p_2, p_3, p_4) \in \mathbb{R}^4$	$\omega \in \mathbb{R}^3$
$SO(2)$	$(\cos \theta, \sin \theta) \in \mathbb{R}^2$	$\dot{\theta} \in \mathbb{R}$

$\mathbf{q}$ Lie group element, $\mathbf{v}$ velocity

- ▶ $\exp \mathbf{v}$: element reached after moving with velocity $\mathbf{v}$ during unit time from origin.
- ▶ $\log \mathbf{q}$ velocity that moves origin to $\mathbf{q}$ in unit time.

Motions on Lie groups

$$\mathbf{q} = \begin{pmatrix} \mathbb{R}^6 & \left\{ \begin{array}{c} q_1 \\ \vdots \\ q_6 \end{array} \right\} \\ \mathbb{R}^6 & \left\{ \begin{array}{c} q_7 \\ \vdots \\ q_{12} \end{array} \right\} \\ SE(3) & \left\{ \begin{array}{c} x \\ y \\ z \\ p_1 \\ p_2 \\ p_3 \\ p_4 \end{array} \right\} \end{pmatrix}$$

$$\dot{\mathbf{q}} = \begin{pmatrix} \mathbb{R}^6 & \left\{ \begin{array}{c} \dot{q}_1 \\ \vdots \\ \dot{q}_6 \end{array} \right\} \\ \mathbb{R}^6 & \left\{ \begin{array}{c} \dot{q}_7 \\ \vdots \\ \dot{q}_{12} \end{array} \right\} \\ \mathbb{R}^6 & \left\{ \begin{array}{c} \mathbf{v} \\ \omega \end{array} \right\} \end{pmatrix}$$

Motions on Lie groups

exp and log are applied to each elementary Lie group.

$$\mathbf{q} = \left(\begin{array}{c} \mathbb{R}^6 \\ \mathbb{R}^6 \\ SE(3) \end{array} \right\{ \begin{array}{c} q_1 \\ \vdots \\ q_6 \\ q_7 \\ \vdots \\ q_{12} \\ x \\ y \\ z \\ p_1 \\ p_2 \\ p_3 \\ p_4 \end{array} \right)$$

$$\log \mathbf{q} = \left(\begin{array}{c} q_1 \\ \vdots \\ q_6 \\ q_7 \\ \vdots \\ q_{12} \\ \log \left(\begin{array}{c} x \\ y \\ z \\ p_1 \\ p_2 \\ p_3 \\ p_4 \end{array} \right) \end{array} \right)$$

Motions on Lie groups

$$\dot{\mathbf{q}} = \begin{pmatrix} \mathbb{R}^6 & \left\{ \begin{array}{c} \dot{q}_1 \\ \vdots \\ \dot{q}_6 \end{array} \right\} \\ \mathbb{R}^6 & \left\{ \begin{array}{c} \dot{q}_7 \\ \vdots \\ \dot{q}_{12} \end{array} \right\} \\ \mathbb{R}^6 & \left\{ \begin{array}{c} \mathbf{v} \\ \omega \end{array} \right\} \end{pmatrix} \quad \exp \dot{\mathbf{q}} = \begin{pmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_6 \\ \dot{q}_7 \\ \vdots \\ \dot{q}_{12} \\ \exp \begin{pmatrix} \mathbf{v} \\ \omega \end{pmatrix} \end{pmatrix}$$

Motions on Lie groups

- ▶ $\mathbf{q} \oplus \dot{\mathbf{q}} \triangleq \mathbf{q} \cdot \exp \dot{\mathbf{q}} \in \mathcal{C}$, $\mathbf{q} \in \mathcal{C}$, $\dot{\mathbf{q}} \in \mathcal{T}_{\mathcal{C}}$,
 - ▶ configuration reached from $\mathbf{q}$ after following the constant velocity $\dot{\mathbf{q}}$ during a unit time,
- ▶ $\mathbf{q}_2 \ominus \mathbf{q}_1 \triangleq \log(\mathbf{q}_1^{-1} \cdot \mathbf{q}_2) \in \mathcal{T}_{\mathcal{C}}$, $\mathbf{q}_1 \in \mathcal{C}$, $\mathbf{q}_2 \in \mathcal{C}$,
 - ▶ constant velocity that moves $\mathbf{q}_1$ to $\mathbf{q}_2$ in unit time.
- ▶ $\mathbf{q}_1 + t(\mathbf{q}_2 \ominus \mathbf{q}_1)$
 - ▶ linear interpolation between $\mathbf{q}_1$ and $\mathbf{q}_2$.

Resources

- ▶ F. Lamiraux and J. Mirabel, Prehensile Manipulation Planning: Modeling, Algorithms and Implementation, IEEE transactions on Robotics, 2022,
- ▶ Online documentation